

Hao Zheng

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Research Profile

I develop multimodal perception, video-based action understanding, and embodied AI methods that enable robots to understand human activities, collaborate effectively with human operators, and improve manufacturing quality and productivity. My work connects vision-language models, video understanding, human-robot collaboration, and human-centric manufacturing.

Research interests: Embodied AI; multimodal and vision-language models; video understanding; human-robot collaboration; human-centric manufacturing; Industry 5.0; intelligent assembly.

Academic Appointments

Postdoctoral Fellow, *Department of Civil Engineering, The University of Hong Kong* Apr 2026 – Present
Advisor: Prof. Xiao Li. Collaborative embodied AI, human-robot collaboration, and digital twins for modular construction.

Postdoctoral Fellow, *Centre for Artificial Intelligence and Robotics / Computer Science and Engineering, New York University Abu Dhabi* May 2025 – Apr 2026
Advisor: Prof. Tuka Alhanai. Multimodal vision-language learning and diffusion-based dual-hand action segmentation for complex assembly videos.

Research Associate, *Center for Transformative Garment Production, The University of Hong Kong* Feb 2025 – May 2025
Advisor: Prof. Hatice Gunes, University of Cambridge. Video- and EMG-based worker-state estimation and human-centric collaboration for garment production.

Education

PhD in Mechanical Engineering, The University of Auckland, New Zealand 2020 – 2025 (*Official award confirmation received on 18 December 2025; hard-copy certificate issued in May 2026.*)
Advisors: Prof. Xun Xu and Dr. Jan Polzer

MEng in Mechanical Design and Theory, China University of Mining and Technology, China 2017 – 2020
Advisor: Prof. Gang Cheng

BEng in Mechanical Engineering and Automation, Yancheng Institute of Technology, China 2013 – 2017

Selected Research Contributions

- **Multimodal embodied intelligence for industrial work.** Developed vision-language and diffusion-based methods for fine-grained dual-hand assembly-action understanding, providing perception capabilities relevant to human-robot collaboration and embodied AI.
- **Assembly datasets and human-robot collaboration.** Created HA-ViD, a large-scale human assembly video dataset, and developed a series of dual-hand action understanding methods. Integrated these advances into a real-time collaborative assembly system with a quality-checking function.
- **Human-centric manufacturing and reliable industrial AI.** Developed video-based fatigue estimation for adaptive task allocation, contributed to Industry 5.0 and human digital-twin research, and designed fault-diagnosis and hazardous-area sensing methods to support safe production.

Publications

First-author, co-first-author*

- Zheng, H.**, Huang, J., Xue, J., Li, X., and Xu, X. (2026). A compositional dual-hand action recognition method for LLM-driven assembly assistance. *Journal of Manufacturing Systems*. (Q1, IF=14.9)
- Zheng, H.**, Wang, H., ..., and Alhanai, T. (2026). Polyphony: diffusion-based dual-hand action segmentation with alternating vision transformer and semantic conditioning. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Zheng, H.**, Geng, Z., and Xu, X. (2026). Foundation model-based generative AI for smart manufacturing: A paradigm shift. *Engineering Management*. (Q1, IF=9.8)
- Zheng, H.**, Huang, J., Zheng, T., Xu, X., and Alhanai, T. (2026). Compositional Context Fine-Tuning Vision-Language Model for Complex Assembly Action Understanding from Videos. *IEEE International Conference on Robotics and Automation (ICRA)*.
- Zheng, H.**, Xia, W., and Xu, X. (2025). A human-robot collaborative assembly framework with quality checking based on real-time dual-hand action segmentation. *Robotics and Computer-Integrated Manufacturing*, 94, 102976. (Q1, IF=12.3)
- Zheng, H.**, Lee, R., and Lu, Y. (2023). HA-ViD: A Human Assembly Video Dataset for Comprehensive Assembly Knowledge Understanding. *Advances in Neural Information Processing Systems (NeurIPS)*, 36, 67069–67081.
- Huang, J. *, **Zheng, H.** *, Ren, Y., Wan, W., Xu, X., and Polzer, J. (2027). Video2Knowledge: Extracting temporally consistent task knowledge from monocular video for robot skill learning. *Robotics and Computer-Integrated Manufacturing*. (Q1, IF=12.3)
- Zheng, H.**, Lee, R., Liang, H., Lu, Y., and Xu, X. (2024). DuCAS: a knowledge-enhanced dual-hand compositional action segmentation method for human-robot collaborative assembly. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 7175–7180.
- Lee, R. K. J. *, **Zheng, H.** *, and Lu, Y. (2024). Human-robot shared assembly taxonomy: A step toward seamless human-robot knowledge transfer. *Robotics and Computer-Integrated Manufacturing*, 86, 102686. (Q1, IF=12.3)
- Zheng, H.**, Lee, R., Lu, Y., and Xu, X. (2024). DuHa: a dual-hand action segmentation method for human-robot collaborative assembly. *IEEE International Conference on Automation Science and Engineering (CASE)*, 522–527.
- Zheng, H.**, Chand, S., Keshvarparast, A., Battini, D., and Lu, Y. (2023). Video-based fatigue estimation for human-robot task allocation optimisation. *IEEE International Conference on Automation Science and Engineering (CASE)*, 1–6.
- Zheng, H.**, Cheng, G., Lu, Y., Liu, C., and Li, Y. (2022). A general fault diagnosis framework for rotating machinery and its flexible application example. *Measurement*, 199, 111497. (Q1, IF=6.1)
- Zheng, H.**, Cheng, G., Li, Y., and Liu, C. (2020). A fault diagnosis method for planetary gear under multi-operating conditions based on adaptive extended bag-of-words model. *Measurement*, 156, 107593. (Q1, IF=6.1)
- Zheng, H.**, Cheng, G., Li, Y., and Liu, C. (2019). A new fault diagnosis method for planetary gear based on image feature extraction and bag-of-words model. *Measurement*, 145, 1–13. (Q1, IF=6.1)
- Wang, S., **Zheng, H.** *, Tang, L., Li, Z., Zhao, R., Lu, Y., and Aw, K. C. (2023). Vibration-based and computer vision-aided nondestructive health condition evaluation of rail track structures. *Journal of Civil Structural Health Monitoring*, 13(1), 1–14. (Q1, IF=5.2)

Selected collaborative publications

- Lu, Y., **Zheng, H.**, Chand, S., Xia, W., Liu, Z., Xu, X., et al., and Bao, J. (2022). Outlook on human-centric manufacturing towards Industry 5.0. *Journal of Manufacturing Systems*, 62, 612–627. (Q1, IF=14.9)
- Chand, S., **Zheng, H.**, and Lu, Y. (2024). A vision-enabled fatigue-sensitive human digital twin towards human-centric human-robot collaboration. *Journal of Manufacturing Systems*, 77, 432–445. (Q1, IF=14.9)
- Xia, W., **Zheng, H.**, Xu, W., and Xu, X. (2025). Large vision-language models enabled novel objects 6D pose estimation for human-robot collaboration. *Robotics and Computer-Integrated Manufacturing*, 95, 103030. (Q1, IF=12.3)
- Ge, C., Yu, X., **Zheng, H.**, Fan, Z., Muhammad, U., Chen, J., and Shum, P. P. (2025). ESC-DRKD: Enhanced skip connection-based direct reverse knowledge distillation for medical image anomaly detection. *Neurocomputing*, 651, 130994. (Q1, IF=6.7)
- Ge, C., Yu, X., **Zheng, H.**, Fan, Z., Chen, J., and Shum, P. P. (2025). A dual reverse distillation scheme for image anomaly detection. *Neurocomputing*, 624, 129479. (Q1, IF=6.7)

Selected Research Projects

Dynamic Coupling Characteristics and Collaborative Control of Large Mirror Manufacturing Equipment

National Natural Science Foundation of China; researcher 2017 – 2020

Developed multimodal fault-diagnosis methods for condition monitoring and operational reliability of rotating machinery.

Intelligent Sensing for Crusher Front-End Protection

School-enterprise collaboration; principal researcher 2018 – 2019

Developed a distributed laser-sensing system for hazardous-area personnel protection.

Intelligent Monitoring and Warning System for Dangerous Areas

School-enterprise collaboration; principal researcher 2019

Developed a computer-vision safety warning system for industrial and mining environments.

Patents

Granted Chinese invention patents

A front-end personnel protection system of mine crusher based on image recognition. CN201910915839.0. Granted Jun 2021.

A protection system and method for front end personnel of underground crusher. CN201811197496.0. Granted Apr 2021.

Teaching and Mentoring

Teaching Assistant, The University of Auckland 2023 – 2024

Supported engineering and manufacturing coursework and student projects.

Academic Service

Conference Reviewer: NeurIPS (2024–2026); International Conference on Automation Science and Engineering (2024–2026).

Journal Reviewer: Robotics and Computer-Integrated Manufacturing; Journal of Manufacturing Systems; IEEE Transactions on Instrumentation and Measurement; Neurocomputing.

Session Chair: Conference on Frontiers of Digital Twin in Intelligent Manufacturing and Smart Cities, 2025.

Honors and Awards

Chinese Government Scholarship (CSC) 2020

National Scholarship for Postgraduate Students 2019

Excellence Student Award, China University of Mining and Technology 2019

Second Prize, May Day Mathematical Modeling Competition 2019